Lesson 1.1

Undefined & Defined Terms

Point, line, plane — and the symbols that go with them

Three words we never define

Every word in geometry is built from three **undefined terms**.



Point A

a location — no size

the pin on a map

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Line AB

straight, goes on forever

both ways: two arrows

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straight, goes on forever

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Plane P

flat surface, forever

a floor that never ends

Defined terms — and their symbols

Line



$\overleftrightarrow{AB}$

two arrows: goes on forever both ways

Defined terms — and their symbols

Line



$\overleftrightarrow{AB}$

two arrows: goes on forever both ways

Ray



$\overrightarrow{AB}$

starts at A — endpoint written **first**

Defined terms — and their symbols

Line



$\overleftrightarrow{AB}$

two arrows: goes on forever both ways

Ray



$\overrightarrow{AB}$

starts at A — endpoint written **first**

Segment



$\overline{AB}$

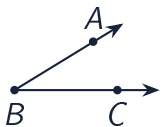
two endpoints — can be measured

Length and angle

Length a number, like 5 cm AB no bar = the **length** of $\overline{AB}$

Length and angle

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Angle 

$\angle ABC$ vertex B goes in the **middle**

The mistake almost everyone makes



$\overrightarrow{AB}$ starts at A



$\overrightarrow{BA}$ starts at B

The mistake almost everyone makes



$\overrightarrow{AB}$ starts at A



$\overrightarrow{BA}$ starts at B

Watch out! $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are **different** rays — they start at different points.

The mistake almost everyone makes



$\overrightarrow{AB}$ starts at A



$\overrightarrow{BA}$ starts at B

Watch out! $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are **different** rays — they start at different points.

But $\overline{AB}$ and $\overline{BA}$ are the **same** segment.

Collinear, coplanar, intersection

- **Collinear**: points on the **same line**.

Collinear, coplanar, intersection

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- **Coplanar**: points on the **same plane**.

Collinear, coplanar, intersection

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- Two lines **intersect** in a **point**.

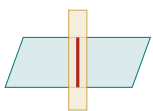


Collinear, coplanar, intersection

- **Collinear**: points on the **same line**.
- **Coplanar**: points on the **same plane**.
- Two lines **intersect** in a **point**.



- Two planes **intersect** in a **line**.



Your turn! Pause the video and try it.

Name this figure with the right symbol.



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Answer: $\overrightarrow{PQ}$ a ray — endpoint P first

Remember

- **Line** $\overleftrightarrow{AB}$: two arrows. **Ray** $\overrightarrow{AB}$: one arrow, endpoint first.
- **Segment** $\overline{AB}$: a bar. **Length** AB: no bar — a number.
- **Angle** $\angle ABC$: the vertex in the middle.

Now practice at your level:

fangmath.com → Geometry → Chapter 1